eirenex

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CHAPTER ONE

INTRODUCTION

Welcome to the EireneX documentation. EireneX is a post-processing package for stand-alone EIRENE simulations. It is designed to work with only the normal input file, an output file created from piping the numerical output of a run, and the data output files defined in block 11 (see getting started). The data types that are read are set in block 11 of the input file. The goal of the developer is to eventually be able to handle every output data type, though only limited number of data types are currently supported. The prettier gif and plotting features also only work for cylindrical geometry, but will soon be available for rectangular and eventually toroidal geometries. The package leverages the 'data array' and 'data set' data types created with the xarray python package. Users will benefit the most if they have a basic understanding of how to manipulate these two data types.

1.1 Supported Data Types

- **Volume Averaged Data**

  -Plasma Species (and Background) Temperature (eV)

  - Block Average Data
  - 1-D X(Radial)–((Averaged in Y(Poloidal) and Z(Toroidal))
  - 1-D Y(Poloidal)–((Averaged in X(Radial) and Z(Toroidal))
  - 1-D Z(Toroidal)–((Averaged in Y(Poloidal) and X(Radial))
  - 2-D X(Radial)- Across Y(Poloidal)-((Averaged in Z(Toroidal))
  - 2-D X(Radial)- Across Z(Toroidal)-((Averaged in Y(Poloidal))
  - 2-D Y(Poloidal)- Across Z(Toroidal)-((Averaged in X(Radial))
  - 3-D Read from piped output data
  - 3-D Read from block 11 fort files

  -Plasma Species (and Background) Density (cm-3)

  - Block Average Data
  - 1-D X(Radial)–((Averaged in Y(Poloidal) and Z(Toroidal))
  - 1-D Y(Poloidal)–((Averaged in X(Radial) and Z(Toroidal))
  - 1-D Z(Toroidal)–((Averaged in Y(Poloidal) and X(Radial))
  - 2-D X(Radial)- Across Y(Poloidal)-((Averaged in Z(Toroidal))
  - 2-D X(Radial)- Across Z(Toroidal)-((Averaged in Y(Poloidal))
  - 2-D Y(Poloidal)- Across Z(Toroidal)-((Averaged in X(Radial))

  - 3-D Read from piped output data
  - 3-D Read from block 11 fort files

-Atom Species Density (cm-3)

  - Block Average Data
  - 1-D X(Radial)–((Averaged in Y(Poloidal) and Z(Toroidal))
  - 1-D Y(Poloidal)–((Averaged in X(Radial) and Z(Toroidal))
  - 1-D Z(Toroidal)–((Averaged in Y(Poloidal) and X(Radial))
  - 2-D X(Radial)- Across Y(Poloidal)-((Averaged in Z(Toroidal))
  - 2-D X(Radial)- Across Z(Toroidal)-((Averaged in Y(Poloidal))
  - 2-D Y(Poloidal)- Across Z(Toroidal)-((Averaged in X(Radial))
  - 3-D Read from piped output data
  - 3-D Read from block 11 fort files

-Molecule Species Density (cm-3)

  - Block Average Data
  - 1-D X(Radial)–((Averaged in Y(Poloidal) and Z(Toroidal))
  - 1-D Y(Poloidal)–((Averaged in X(Radial) and Z(Toroidal))
  - 1-D Z(Toroidal)–((Averaged in Y(Poloidal) and X(Radial))
  - 2-D X(Radial)- Across Y(Poloidal)-((Averaged in Z(Toroidal))
  - 2-D X(Radial)- Across Z(Toroidal)-((Averaged in Y(Poloidal))
  - 2-D Y(Poloidal)- Across Z(Toroidal)-((Averaged in X(Radial))
  - 3-D Read from piped output data
  - 3-D Read from block 11 fort files

-Test Ion Species Density (cm-3)

  - Block Average Data
  - 1-D X(Radial)–((Averaged in Y(Poloidal) and Z(Toroidal))
  - 1-D Y(Poloidal)–((Averaged in X(Radial) and Z(Toroidal))
  - 1-D Z(Toroidal)–((Averaged in Y(Poloidal) and X(Radial))
  - 2-D X(Radial)- Across Y(Poloidal)-((Averaged in Z(Toroidal))
  - 2-D X(Radial)- Across Z(Toroidal)-((Averaged in Y(Poloidal))
  - 2-D Y(Poloidal)- Across Z(Toroidal)-((Averaged in X(Radial))
  - 3-D Read from piped output data
  - 3-D Read from block 11 fort files

- Surface Averaged Data

1.2 Contact Information

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1.3 Getting Started

The following will show how to get started using eirenex

1.3.1 Download

First, you will need to download or clone the repository from:

https://github.com/nbb2

Once you have it downloaded, put yourself into the main directory:

```
>>> cd eirenex
```

We need to make sure that you have all of the needed modules for eirenex to run. You can find them in the requirements.txt file. Now, in the command line run:

```
>>> pip install .
```

This should download all of the required packages. If this doesn't work, individually download the needed packages. Next, we will run the test scripts to make sure that everything is working alright. These test scripts should be used consistently and updated as the code is modified. Pytest will be needed in order to run the following command:

```
>>> pytest
```

If all of the test pass, then we can now move on to the first example.

1.3.2 Eirene Input File Formatting

In order for the full range of date to be read by EireneX, the input Eirene input file needs to be modified in Block 11 so that the desired data types are saved in newly created files with the name fort.XX. An example can be seen below. In order for the batching scripts to work correctly, plasma temperature data goes to fort.90, plasma denisty data goes to fort.91, atom density data goes to fort.92, molecule density data goes to fort.93, and test ion denisty data goes to fort.94.

```
*** 11. Data for numerical and graphical output
TFFFT TFFTT FFTFF TTTTT FFFFF FFFFF FFFF
TFFF
     5     0
    -2     3     0     0    90
    -4     3     0     0    91
     1     3     0     0    92
     2     3     0     0    93
     3     3     0     0    94
```

CHAPTER

TWO

MODULES AND FUNCTIONS

2.1 eirenex.data

`data.run(FILE_in, FILE_out, Plasma_Temp_Data, Plasma_Dens_Data, Atom_Dens_Data, Mol_Dens_Data, Ion_Dens_Data, DIRECTORY)`

Create an xarray object of data from a single EIRENE run

Parameters

- **FILE_in** (*str*) – The standard input for an EIRENE standalone run
- **FILE.out** (*str*) – The standard output for an EIRENE standalone run

:param str Plasma_Temp_Data fort.90 file of plasma temnperature data :param str Plasma_Dens_Data fort.91 file of plasma density data :param str Atom_Dens_Data fort.92 file of atom density data :param str Mol_Dens_Data fort.93 file of molecule denisty data :param str Ion_Dens_Data fort.94 file of test ion denisty data :param str DIRECTORY: The directory where the output dataset should be saved :return: An xarray dataset with data from the EIRENE output :rtype: xarray dataset

`data.batch(DS_LIST, BATCH_NAME, DIRECTORY)`

Create an xarray object of data from a set of xarray datasets

Parameters

- **DS_LIST** (*file*) – A list of Dataset objects to be combined into a single dataset
- **BATCH_NAME** (*str*) – A name for the batch
- **DIRECTORY** (*str*) – A file path where the new dataset should be saved

Returns An xarray dataset with data from a batch of EIRENE output

Return type xarray dataset

`data.auto(DIRECTORY, BATCH_NAME)`

Create an xarray objects for each EIRENE run in a directory, and a combined dataset of all of the individual datasets

Parameters

- **DIRECTORY** (*str*) – The directory containing a set of EIRENE input and output files
- **BATCH_NAME** (*str*) – A name for the batch
- **DIRECTORY** – A file path where the new dataset should be saved

Returns NONE

Return type NONE

2.2 eirenex.plot

`plot.avg_t_pls(DS, SPC)`

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.avg_d_pls(DS)`

Plots data from a Dataset for a specific species

param xarray dataset DS Dataset of EIRENE output data

param str SPC The species to be plotted

return fig1,fig2

rtype matplotlib figures

`avg_d_atm(DS)`

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.avg_d_mol(DS)`

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_1d_t_pls(DS)`

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_1d_d_pls(DS)`

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_1d_d_atm(DS)`
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_1d_d_mol(DS)`
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_1d_d_ion(DS)`
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_2d_t_pls(DS)`
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.cyl_2d_d_pls(DS)`
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`cyl_2d_d_atm`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`cyl_2d_d_mol`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`cyl_2d_d_ion`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_PLS_1D_TEMP`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_PLS_1D_DENS`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_ATM_1D_DENS`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*`xarray`*) – Dataset of EIRENE output data
- **`SPC`** (*`str`*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_MOL_1D_DENS`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*`xarray`*) – Dataset of EIRENE output data
- **`SPC`** (*`str`*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_ION_1D_DENS`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*`xarray`*) – Dataset of EIRENE output data
- **`SPC`** (*`str`*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_PLS_1D_TEMP`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*`xarray`*) – Dataset of EIRENE output data
- **`SPC`** (*`str`*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_PLS_1D_DENS`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*`xarray`*) – Dataset of EIRENE output data
- **`SPC`** (*`str`*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_ATM_1D_DENS`**(*DS*)

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

plot.**gif_TIME_STP_MOL_1D_DENS**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

plot.**gif_TIME_STP_ION_1D_DENS**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

plot.**gif_ITERATION_PLS_2D_TEMP**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

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- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

plot.**gif_ITERATION_ATM_2D_DENS**(*DS*)
Plots data from a Dataset for a specific species

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- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_MOL_2D_DENS`**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*xarray*) – Dataset of EIRENE output data
- **`SPC`** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_ITERATION_ION_2D_DENS`**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*xarray*) – Dataset of EIRENE output data
- **`SPC`** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_PLS_2D_TEMP`**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*xarray*) – Dataset of EIRENE output data
- **`SPC`** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_PLS_2D_DENS`**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*xarray*) – Dataset of EIRENE output data
- **`SPC`** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_ATM_2D_DENS`**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **`dataset DS`** (*xarray*) – Dataset of EIRENE output data
- **`SPC`** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.`**`gif_TIME_STP_MOL_2D_DENS`**(*DS*)
Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

`plot.gif_TIME_STP_ION_2D_DENS(DS)`

Plots data from a Dataset for a specific species

Parameters

- **dataset DS** (*xarray*) – Dataset of EIRENE output data
- **SPC** (*str*) – The species to be plotted

Returns fig1,fig2

Return type matplotlib figures

2.3 eirenex.geometry

`geometry.AUG_TOY_GEOM()`

Builds needed datatypes for plotting using a toy AUG geometry

Returns aug_toy_geom

Return type xarray Dataset

2.4 eirenex.intern

Contains functions used by the code, but should not be called by the user.

CHAPTER THREE

EXAMPLES

The current version of eirenex as three examples using data from an EIRENE standalone

3.1 Working in IMAS Environment

If working in the IMAS environment, make sure that the version of python being used also has matching 'xarray' and 'matplolib' modules loaded. For example

[bartlen@hpc-login02 test_1_raw]$ module purge

[bartlen@hpc-login02 test_1_raw]$ module load Python/3.6.4-intel-2018a

[bartlen@hpc-login02 test_1_raw]$ module load matplotlib/2.1.2-intel-2018a-Python-3.6.4

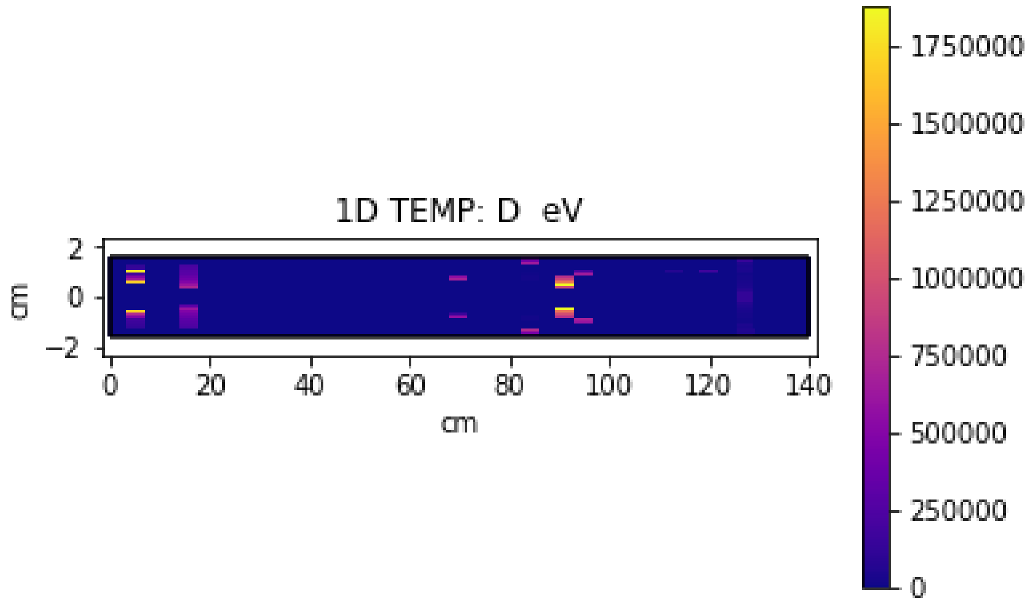
[bartlen@hpc-login02 test_1_raw]$ module load xarray/0.10.8-intel-2018a-Python-3.6.4

[bartlen@hpc-login02 test_1_raw]$ python

Python 3.6.4 (default, Dec 12 2019, 22:32:08) [GCC Intel(R) C++ gcc 6.4 mode] on linux Type "help", "copyright", "credits" or "license" for more information.

```
>>> import eirenex as EX
>>> DS = EX.data.run('./examples/cylinder/test_3D_raw/TEST_3D.in','./examples/
↪cylinder/test_3D_raw/TEST_3D.out')
```

3.2 Cylinder



First, we will begin by just working with 0D (Block Average) data and 1D data. The directory:

```
>>> examples/cylindrical/
```

contains three examples each split into a 'raw' and 'complete' subfolders. After the examples are complete, the 'raw' folders should be identical to the 'complete' folders.

We will begin by using the main function of EireneX: eirenex.data.run(FILE_in,FILE_out). This function takes two arguments, an EIRENE input file, and the piped standard output of the EIRENE run with that file. When these are passed into it, an xarray Dataset is returned and saved into the directory it is run in. You should look up all of the methods that can be used with xarray datasets at

> http://xarray.pydata.org/en/stable/#

Lets make the dataset, while in the run directory, enter into a python session and import eirenex:

```
>>> python
```

```
>>> import eirenex as EX
```

Now, use the 'run function' on the files in test_1d.

```
>>> DS = EX.data.run('examples/cylinder/test_1_raw/test_1.in','examples/cylinder/test_
↪1_raw/test_1.out')
>>> DS
```

Note the aspects of the dataset that has been created. We can quickly access aspects of the Dataset using the isel method:

```
>>> DS.PLS_BLA_TEMP.isel(ITERATION = 0 ,PLS_SPECIES = 0).data
```

We have selected to display the data first iteration and first species. We can quickly plot this data using:

```
>>> import matplotlib.pyplot as plt
>>> plt.show(DS.PLS_BLA_TEMP.isel(ITERATION = 0 ,PLS_SPECIES = 0).plot(x = 'TIME_STP
↪'))
```

2D plots can also be quickly generated using Dataset methods;

```
>>> plt.show(DS.PLS_XR_TEMP.isel(ITERATION =0 ,PLS_SPECIES = 1).plot())
```

In the plot module of Eirenex, there are several plotting functions that are specific to the cylindrical geometry:

```
>>> plt.show(EX.plot.cyl_1d_t_pls(DS,'D+'))
```

Now, lets use EX.run on the second test to see some methods that work on 2D data:

```
>>> DS = EX.data.run('examples/cylinder/test_2D_raw/test_2D.in','examples/cylinder/
↪test_2D_raw/test_2D.out')
>>> plt.show(EX.plot.cyl_2d_d_atm(DS,'D'))
>>> EX.plot.gif_TIME_STP_ATM_2D_DENS(DS,'D')
```

We have now created a new dataset, a 2D plot, and a gif of the same2D plot across time steps of the run.

Now, lets use run on a file set that has the new fort.9X data formats

```
>>> DS = EX.data.run('examples/cylinder/test_fort/input.dat','examples/cylinder/test_
↪fort/run.log','examples/cylinder/test_fort/fort.90','examples/cylinder/test_fort/
↪fort.91','examples/cylinder/test_fort/fort.92','examples/cylinder/test_fort/fort.93
↪','examples/cylinder/test_fort/fort.94','examples/cylinder/test_fort/')
>>> DS
```

Note how we still have read the data from the input and output file, but now we are also reading the data from the individual data files and the resulting dataset has more information.

Finally, lets use the eirenex.data.auto() function to create a create a dataset composed of smaller datasets of a batch of EIRENE runs:

```
>>> EX.data.auto('examples/cylinder/test_batch/','New_Test_Batch_')
>>> import xarray as xr
>>> DSB = xr.load_dataset('examples/cylinder/test_batch/New_test_Batch_EIR_XBATCH_OUT.
↪nc')
>>> DSB
```

Note, now each individual run is a part of the BATCHID Dimension.

CHAPTER

FOUR

INDICES AND TABLES

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INDEX